

BRAIN INFECTIONS

Infection of brain tissue or its protective layers can be caused by a variety of viruses, bacteria, and tropical parasites. Infection of the brain, or encephalitis, is a rare complication of a viral infection, such as mumps or measles. It can be fatal, with babies and elderly people being most at risk.

MENINGITIS

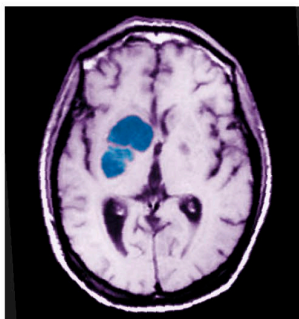
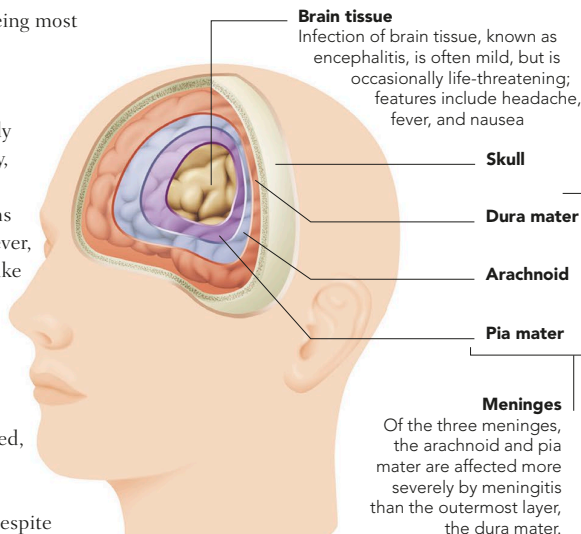
Inflammation of the meninges is usually caused by a virus or bacterium. Initially, meningitis may cause vague flu-like symptoms. More pronounced symptoms may also develop, such as headache, fever, nausea, vomiting, stiff neck, and a dislike of bright light. In meningitis due to *Meningococcus* bacteria, there is a distinctive reddish-purple rash. If meningitis is suspected, immediate admission to hospital is necessary for tests. If bacterial meningitis is confirmed, treatment in intensive care is often required, and a complete recovery may take weeks or months. It can be fatal despite treatment. Recovery from viral meningitis usually takes up to two weeks. No specific treatment is needed.

BRAIN ABSCESS

An abscess is a collection of pus. Brain abscesses are rare, and are usually caused by bacteria that have spread to the brain from an infection in tissues in the skull. Treatment consists of high doses of antibiotics and possibly corticosteroids to control swelling of the brain. Surgery may be needed to drain pus through a hole drilled in the skull. If given early treatment, many people recover from it. However, some have persistent problems, such as seizures or slurred speech.

SITES OF INFECTION

Infectious organisms can affect the brain itself, the three membranes (meninges) that surround the brain, or both. Infections can reach the brain through the blood, but can also spread from a nearby infection (such as an ear infection) or through a skull wound.



BRAIN ABSCESS

This MRI scan of the brain shows an abscess (blue area) due to a fungal infection in a person who has AIDS. People with AIDS are at increased risk of developing a brain abscess.



TESTING A MENINGITIS RASH

In meningococcal meningitis, bacteria in the blood may cause dark-red or purple spots that turn into blotches. The rash does not fade when pressed with a glass.